Project Initialization and Planning Phase

Date	19 June 2025
Team ID	SWTID1749908722
Project Title	CardMaster: Intelligent Playing Card Recognition using Transfer Learning
Maximum Marks	3 Marks

Project Proposal (Proposed Solution)

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To develop an intelligent system that accurately identifies playing cards from real-time images using transfer learning techniques, thereby aiding gaming applications, AR systems, and visually impaired users.
Scope	The system will identify **standard 52-card playing cards and a joker card**. The input will be static images. The solution will support recognition in **varying lighting conditions and angles**.
Problem Statement	
Description	In many games, especially online or automated physical games, identifying playing cards via visual input is tedious, error-prone, and lacks automation. Manual tracking or expensive proprietary systems are often required.
Impact	Solving this problem will: **Automate** card recognition in real-world settings. Enable the integration of AI in **AR casinos, educational apps, and accessibility tools**.

Proposed Solution

Approach	Utilize transfer learning to construct the model architecture and conduct initial training to adapt it for the playing card recognition task.
Key Features	Accurate recognition of **card suit and rank** from a single image. **Web interface** for interactive card detection.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU (Google Collaboratory)
Memory	RAM specifications	4 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask, TensorFlow / Keras Kaggle API
Libraries	Additional libraries	scikit-learn, pandas, numpy, IPython.display (Image, display) os
Development Environment	IDE, version control	Google collab , Flask
Data		
Data	Source, size, format	https://www.kaggle.com/datasets/gpiosenka/cards-image-datasetclassification 53 classes 7624 train, 265 test, 265 validation images 224 X 224 X 3 jpg format